MINIPROJECT REPORT

ON

REAL-TIME AUTOMATIC CRACK DETECTION SYSTEM FOR INTERLOCKED METAL CHAINS

Submitted by

AAYUSHI MICHAEL (SJC23ECE002)

ALAN JIBIN (SJC23ECE011)

DOMINIC GEORGE (SJC23ECE036)

YADUKRISHNA HARI (SJC23ECE072)

to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

Electronics and Computer Engineering

Department of Electronics and Computer Engineering

St. Joseph's College of Engineering and Technology, Palai

(Autonomous)

MAR : 2026

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(AUTONOMOUS)

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

CERTIFICATE

This is to certify that the report entitled **"Real-Time Automatic Crack Detection System For Interlocked Metal Chains "** submitted by **Aayushi Michael (SJC23ECE002), Alan Jibin (SJC23ECE011), Dominic George (SJC23ECE036), and Yadukrishna Hari (SJC23ECE072)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Computer Engineering is a bonafide record of the miniproject carried out by them under my guidance and supervision.

Project Guide
Prof.Al-Saheer S.S
Assistant Professor
Department of ER

Project Coordinator
Prof. Soya Treesa Jose
Assistant Professor
Department of ER

Place : Choondacherry
Date :

Head of the Department
Dr. Giby Jose
Associate Professor
Department of ER

Acknowledgement

We wish to record our indebtedness and thankfulness to all who helped us complete this Mini Project work titled REAL-TIME AUTOMATIC CRACK DETECTION SYSTEM FOR INTERLOCKED METAL CHAINS

We would like to convey our special gratitude to Dr. V.P. Devassia, Principal, SJCET, Palai, for providing us with all the facilities for doing this project. We express our sincere thankfulness to Dr. Giby jose, Head of the department, Department of Electronics & Computer Engineering for his cooperation and valuable suggestions. Also, we express our sincere thanks to the Mini project co-ordinators Ms. Pinky M S & Ms. Soya Treesa Jose for their helpful feedback and timely assistance. We are especially thankful to our guide, Mr. Al-Saheer S.S, Assistant Professor, Department of Electronics & Computer Engineering for giving us valuable suggestions and critical inputs through guidance and support.

We also extend our thanks to all friends who helped us by giving motivation. Their smallest piece of advice was really valuable. Once again we convey our gratitude to all those persons who had directly or indirectly influenced our work as a whole.

Aayushi Michael

Alan Jibin

Dominic George

Yadukrishna Hari

Abstract

Industrial metal chains are widely used in various mechanical and manufacturing systems, where their reliability is critical for safe and efficient operation. However, these chains are highly susceptible to surface cracks due to continuous stress, wear, and environmental factors. Such cracks, if left undetected, can lead to serious safety hazards, unexpected equipment failure, and significant economic losses. Therefore, ensuring timely and accurate inspection of metal chains is essential in industrial environments.

Traditional manual inspection methods are often inefficient and unreliable, particularly in high-speed or continuous production settings. Human inspection is prone to fatigue, inconsistency, and limited ability to detect very small or early-stage cracks. As a result, minor defects may go unnoticed until they develop into major failures. This limitation highlights the need for an automated and more accurate inspection system.

To address these challenges, the proposed system introduces a real-time, automated approach for detecting surface cracks on metal chains. The system utilizes image processing techniques to analyze images captured from chains moving on a conveyor belt. By employing a non-contact inspection method, the system ensures that the chains are not physically disturbed during the inspection process, thereby maintaining their integrity while achieving consistent performance.

The system is capable of providing immediate visual feedback by highlighting detected cracks in the processed images. In addition, it records inspection data for further analysis and documentation. This combination of real-time monitoring and data storage enhances the overall effectiveness of the inspection process. By improving detection accuracy, reducing human dependency, and ensuring better quality control, the proposed system offers a reliable solution for modern industrial applications.

Table of Contents

List of Figures

Chapter 1

Introduction

The project titled Real-Time Automatic Crack Detection System for Interlocked Metal Chains is aimed at addressing a critical challenge in modern industrial environments—reliable and efficient detection of surface defects in metal chains. Metal chains are widely used in a variety of industrial applications such as conveyor systems, lifting equipment, hoisting mechanisms, and heavy machinery. These chains operate under continuous mechanical stress, heavy loads, and harsh environmental conditions, which makes them highly vulnerable to wear and tear over time. One of the most common and dangerous defects that develop in such chains is surface cracking. If not detected at an early stage, these cracks can propagate and eventually lead to sudden chain failure, resulting in severe safety hazards, equipment damage, and costly downtime in production processes.

Traditionally, inspection of metal chains is carried out through manual visual inspection methods. While this approach is simple and does not require sophisticated equipment, it suffers from several limitations. Manual inspection is time-consuming, labor-intensive, and highly dependent on the skill and experience of the operator. Moreover, it is often ineffective in identifying small or early-stage cracks, especially when chains are moving at high speeds in production lines. Inconsistent inspection results and human fatigue further reduce the reliability of this method. As industries continue to adopt automation and high-speed manufacturing practices, the need for a more accurate, consistent, and real-time inspection system has become increasingly important.

To overcome these limitations, the proposed system introduces an automated and non-contact crack detection mechanism using image processing and computer vision techniques. The system captures high-resolution images of metal chains as they move along a conveyor and processes these images in real time to identify defects. By automating the inspection process, the system eliminates human error, improves detection accuracy, and ensures continuous monitoring without interrupting production. This approach not only enhances industrial safety but also contributes to improved product quality and operational efficiency.

1.1 Motivation

The motivation for developing this project stems from the increasing demand for safer, more reliable, and efficient industrial operations. Metal chains are critical components in many industrial systems, and their failure can have serious consequences. A single undetected crack can lead to catastrophic failure, causing accidents, injuries, and even loss of life in extreme cases. Additionally, such failures can result in significant financial losses due to equipment damage, production downtime, and costly repairs. These risks highlight the importance of early and accurate detection of defects in metal chains.

Despite the importance of chain inspection, existing methods are largely inadequate. Manual inspection, which is the most commonly used approach, is not only inefficient but also unreliable in detecting small defects. Inspectors may overlook tiny cracks due to fatigue, poor lighting conditions, or the high speed of moving chains. Furthermore, manual inspection cannot keep up with the demands of modern automated production lines, where components move rapidly and continuously. This creates a gap between the need for high-speed inspection and the capabilities of traditional methods.

Another key motivation is the growing trend toward Industry 4.0 and smart manufacturing, where automation and intelligent systems play a crucial role in improving productivity and quality. Integrating automated inspection systems into production lines allows industries to monitor equipment health in real time and make data-driven decisions. The proposed system aligns with this trend by leveraging computer vision and image processing technologies to create a smart inspection solution. It reduces reliance on human intervention, minimizes errors, and ensures consistent performance.

Additionally, the project is motivated by the need to shift from reactive maintenance to preventive and predictive maintenance strategies. Instead of waiting for a failure to occur, industries can use real-time detection systems to identify potential issues early and take corrective action before they escalate. This not only improves safety but also reduces maintenance costs and extends the lifespan of equipment. By providing accurate and timely information about the condition of metal chains, the proposed system plays a vital role in achieving these goals.

1.2 Objective and Scope

1.2.1 Objective

The primary objective of this project is to design and develop an automated, non-contact crack detection system capable of identifying surface cracks in interlocked metal chains accurately and in real time. The system aims to use image processing techniques to analyze images captured from a camera as the chains move along a conveyor. It focuses on developing a reliable mechanism that can distinguish between cracked and defect-free chains by detecting irregularities on their surface.

Another important objective is to identify small and early-stage cracks that are often missed during manual inspection. The system is designed to provide real-time visual feedback by marking the detected crack locations on the output display, enabling operators to easily recognize defects. It also aims to capture high-quality images of moving chains using a controlled inspection setup to ensure accurate analysis.

In addition, the project seeks to reduce manual effort, minimize human error, and improve inspection speed and efficiency. By automating the inspection process, the system enhances industrial safety and ensures better quality control in manufacturing and maintenance operations.

1.2.2 Scope

The scope of this project includes the development of a real-time automated inspection system for detecting surface cracks in metal chains used in industrial applications. It covers both hardware and software components required for implementing the system. The hardware setup includes an industrial camera, camera lens, lighting system, conveyor belt, frame and mounting structure, processing unit, display unit, power supply, and connecting components.

On the software side, the system uses Python programming and image processing libraries such as OpenCV to perform various operations, including image acquisition, region of interest selection, preprocessing, and edge detection for crack identification. These processes enable the system to analyze chain surfaces and detect defects accurately.

The system is intended for use in industrial environments such as manufacturing plants, conveyor systems, material handling units, and lifting equipment, where continuous monitoring of metal chains is required. It supports applications in quality control, preventive maintenance, and safety monitoring by enabling real-time inspection during operation.

The scope of the project is limited to detecting surface cracks using image processing techniques as described. However, it provides a foundation for future enhancements and improvements in system performance and capabilities.

1.3 Contributions

The Real-Time Automatic Crack Detection System for Interlocked Metal Chains contributes to the field of industrial automation and quality control by providing an efficient and reliable solution for detecting cracks in metal chains. One of the key contributions is the development of an automated inspection system that replaces traditional manual methods with a faster and more accurate approach. By using image processing techniques, the system is able to detect cracks in real time, improving consistency and reducing errors.

Another important contribution is the implementation of a non-contact inspection method. The system uses a camera-based setup to capture images without physically interacting with the chains, allowing continuous monitoring without interrupting production processes. This makes it suitable for use in high-speed industrial environments.

The project also contributes by providing real-time visual feedback, where detected cracks are highlighted on the display. This helps operators quickly identify defective chains and take corrective actions. Additionally, the system can store inspection data, which can be used for analysis and improving maintenance strategies.

Furthermore, the system enhances industrial safety by reducing the risk of chain failure and associated hazards. It helps protect workers, machinery, and infrastructure by ensuring early detection of defects. The project also improves product quality by ensuring that only defect-free chains are used in operations.

Overall, the project contributes to improved efficiency, reduced operational costs, enhanced safety, and better quality control in industrial environments.

Chapter 2

Literature Review

The literature review for this project highlights several significant research contributions in the field of metal crack detection and industrial quality control.

Zhang Cheng et.al. in [1] proposed an improved RT-DETR algorithm that enhances feature fusion and coordinate attention mechanisms. This model achieved high detection accuracy along with real-time performance, making it suitable for industrial inspection systems. The emphasis on real-time processing and improved accuracy in this work inspired the idea of developing a system capable of detecting cracks instantly during operation.

Fuqiang Zhou et.al. in [2] presented a lightweight and efficient model optimized for accurate crack detection with reduced computational cost. This work highlighted the importance of achieving a balance between performance and efficiency, especially in real-time industrial environments. It influenced the approach of designing a system that is not only accurate but also efficient enough to operate continuously without excessive computational requirements.

Jie Liu and Zhuoxing Li in [3], focused specifically on defect detection in chain-based systems using deep learning techniques. This research demonstrated the feasibility of implementing automated inspection systems directly in production lines. It strongly influenced the decision to apply the crack detection system to interlocked metal chains and to design the system for continuous monitoring on a conveyor setup.

K. Anitha in [4], emphasized the need for modern quality control systems that reduce dependency on manual inspection. It highlighted the drawbacks of traditional inspection methods, such as being labor-intensive, subjective, and prone to human error. This study reinforced the motivation to develop an automated system that improves consistency, reliability, and efficiency in inspection processes.

Mohammed A. Abou-Khousa in [5], discussed advanced non-destructive testing methods used for detecting cracks in metallic structures. This work highlighted the importance of reliable inspection techniques for maintaining structural integrity. Although it used different technology, it emphasized the need for accurate and dependable crack detection systems, which aligns with the objective of the proposed project.

Based on the insights gained from these studies, the present work evolves as a practical and simplified approach for real-time crack detection using image processing techniques. While advanced models such as RT-DETR and YOLOv5 provide high accuracy, the proposed system focuses on implementing a cost-effective and efficient solution using computer vision methods like image acquisition, preprocessing, edge detection, and crack identification. The idea of real-time inspection, automation, and integration with production lines is directly inspired by the reviewed literature.

Thus, the proposed methodology combines the key concepts of real-time detection, automation, and reliability derived from previous research, while simplifying the implementation using accessible tools such as Python and OpenCV. This evolution results in a system that is suitable for industrial applications, improves inspection accuracy, reduces manual effort, and ensures continuous monitoring of metal chains.

2.1 System Description

The proposed system is a real-time automated crack detection system designed to inspect interlocked metal chains in industrial environments. The system operates based on a non-contact inspection mechanism, where images of metal chains are captured and processed to detect surface cracks. The overall system consists of both hardware and software components working together in a sequential manner. The hardware setup includes an industrial camera, lighting system, conveyor belt arrangement, processing unit, and display unit. The software component is responsible for processing the captured images using image processing techniques to identify defects.

The system is designed to function continuously as the chains move along a conveyor belt. The camera captures live images of the moving chains, which are then sent to the processing unit. The processing unit applies various image processing steps such as preprocessing, edge detection, and crack identification to analyze the chain surface. The final output is displayed in real time, where detected cracks are highlighted for easy identification. This integrated approach ensures efficient and accurate inspection without interrupting the production process.

The system block diagram represents the flow of data and the interaction between different components of the crack detection system. The first block in the system is the image acquisition unit, which consists of an industrial camera positioned above or near the conveyor belt. This camera continuously captures images of the metal chains as they move along the production line. Proper lighting is provided to ensure that the captured images are clear and free from shadows or distortions.

The captured images are then passed to the processing unit, which forms the core of the system. This unit processes the input images using a series of image processing techniques. Initially, the region of interest is selected to focus only on the chain and eliminate unnecessary background information. The image is then preprocessed by converting it into grayscale and improve image quality.The edges are analyzed to detect irregular patterns that indicate the presence of cracks. Once a crack is identified, the system marks the location of the defect on the image.

The final processed output is sent to the display unit, where the results are shown in real time. The output includes images with highlighted crack regions, allowing operators to easily identify defective chains. The system may also generate logs or reports for further analysis. This block-wise flow ensures systematic processing and accurate detection of cracks.

The first step is image acquisition, where the camera captures live images of the metal chains moving on the conveyor belt. This step is crucial as the quality of the captured images directly affects the accuracy of crack detection.

The next step is region of interest selection, where only the portion of the image containing the metal chain is selected for further processing. This helps in removing background noise and focusing the analysis on the relevant area. After this, preprocessing is performed on the selected image. The image is converted into grayscale to simplify processing.

In the crack identification stage, the system analyzes these edges to detect patterns that correspond to cracks. Any discontinuities or irregular shapes in the edges are considered as potential defects.

Finally, the output is displayed on the screen, where detected cracks are marked clearly on the image. This allows operators to quickly identify and take action on defective chains. The entire process is performed in real time, enabling continuous monitoring of the chains during operation.

The central component of the circuit is the processing unit, which is responsible for executing the image processing algorithms. The industrial camera is connected to this processing unit, providing a continuous stream of image data.

A lighting system is integrated into the setup to ensure proper illumination of the metal chains during image capture. This helps in improving image clarity and detection accuracy. The conveyor belt mechanism is used to move the chains through the inspection area at a controlled speed, allowing the camera to capture consistent images.

The display unit is connected to the processing unit to show the output results. It displays the processed images along with highlighted crack regions, enabling real-time monitoring. The system also includes a power supply unit that provides the necessary electrical power to all components, along with cables and connectors to establish proper connections.

The circuit is designed in such a way that all components work in synchronization, ensuring smooth data flow from image acquisition to output display. This integrated hardware setup supports the overall functioning of the crack detection system.

The effectiveness of the proposed system lies in the integration of hardware and software components into a unified framework. The camera, lighting system, and conveyor setup work together to provide high-quality input images, while the processing unit executes the image processing algorithms to detect cracks. The display unit ensures that the results are communicated clearly to the user in real time.

The system operates continuously, making it suitable for industrial environments where uninterrupted monitoring is required. Each component plays a specific role, and their combined operation ensures accurate detection, reduced manual effort, and improved efficiency. The integration of these components forms a complete and functional system capable of addressing the limitations of traditional inspection methods.

2.2 Existing Solutions

Several existing solutions are available for crack detection in metal components and industrial chains. These solutions mainly include non-destructive testing methods, manual inspection methods, and automated vision-based systems. One of the most commonly used existing solutions is manual visual inspection, where workers inspect metal chains and components visually to identify cracks or defects. This method is simple and low cost, but it is time-consuming, less accurate, and small cracks are often missed due to human error. It is also difficult to inspect chains continuously while machines are operating.

Another existing solution is Non-Destructive Testing (NDT) methods such as ultrasonic testing, radiography (X-ray), eddy current testing, and magnetic particle inspection. These methods are highly accurate and can detect both surface and internal cracks. However, these methods are expensive, require specialized equipment and trained personnel, and are not suitable for continuous real-time inspection in production lines.

Recently, automated crack detection systems using image processing and computer vision have been developed. These systems use cameras to capture images of metal surfaces and apply image processing algorithms such as edge detection, thresholding, and crack pattern analysis to detect cracks automatically. These systems are faster, more reliable, and suitable for real-time industrial inspection.

Some modern systems also use machine learning and deep learning techniques such as Convolutional Neural Networks (CNN), YOLO, and other object detection models to identify cracks with higher accuracy. However, these systems require large datasets, powerful hardware, and complex implementation.

Based on the limitations of existing solutions, there is a need for a cost-effective, real-time automated crack detection system, which is the focus of this project.

2.3 Summary

The project titled Real-Time Automatic Crack Detection System for Interlocked Metal Chains presents an efficient and reliable solution to address the challenges associated with detecting surface defects in industrial metal chains. Metal chains are widely used in applications such as conveyor systems, lifting equipment, and heavy machinery, where they are continuously subjected to stress, load, and wear. Over time, these conditions lead to the formation of surface cracks, which, if left undetected, can result in equipment failure, safety hazards, and significant economic losses. Traditional inspection methods mainly rely on manual visual inspection, which is time-consuming, prone to human error, and ineffective in detecting small or early-stage cracks, especially in fast-moving production environments. This highlights the need for an automated system capable of performing accurate and real-time inspection.

The proposed system utilizes a non-contact inspection approach based on image processing and computer vision techniques. An industrial camera is used to capture live images of metal chains as they move along a conveyor belt. These images are then processed using a series of steps, including region of interest selection, grayscale conversion, and noise reduction. By analyzing these irregular patterns, the system is able to identify cracks and mark their locations on the output display. This real-time processing capability ensures continuous monitoring without interrupting the production process, thereby improving efficiency and reliability.

The system is designed with both hardware and software components working in integration. The hardware setup includes an industrial camera, camera lens, lighting system, conveyor belt, processing unit, display unit, and power supply. The software component is developed using Python programming and utilizes libraries such as OpenCV for image processing tasks. The system is capable of capturing high-quality images of moving chains and processing them instantly to provide accurate results. In addition to detecting cracks, the system also generates visual outputs and logs that can be used for further analysis and maintenance planning.

The project aims to achieve several important objectives, including improving inspection accuracy, reducing manual effort, and enabling the detection of small and early-stage cracks that are often missed in traditional methods. It also focuses on providing real-time visual feedback to operators, allowing them to quickly identify and address defects. The system is intended for use by manufacturing plant operators, maintenance engineers, quality control departments, and industrial chain manufacturers, making it applicable across a wide range of industrial settings.

The implementation of this system offers significant benefits in terms of safety, reliability, and efficiency. By enabling early detection of defects, the system helps prevent chain failures and reduces the risk of accidents in industrial environments. It also supports preventive maintenance by allowing timely replacement of damaged chains, thereby minimizing downtime and repair costs. Furthermore, the system improves product quality by ensuring that only defect-free chains are used in operations. Overall, the project contributes to enhancing industrial safety, optimizing inspection processes, and supporting the transition toward automated and intelligent manufacturing systems.

Chapter 3

Proposed Methodology

The proposed methodology of the system is based on a sequence of image processing steps designed to detect surface cracks in metal chains in real time. The process begins with image acquisition, where an industrial camera captures live images of metal chains as they move continuously on a conveyor belt. This step ensures that the system receives real-time visual input without interrupting the production process. The quality of image acquisition is enhanced using a proper lighting system and controlled setup, which helps in capturing clear and detailed images of the chain surface.

Once the image is captured, the next step is the selection of the region of interest. In this stage, only the portion of the image that contains the metal chain is selected for further processing, while unnecessary background elements such as the conveyor belt or surrounding environment are removed. This helps in reducing noise and improving the efficiency and accuracy of subsequent processing steps.

The selected image then undergoes preprocessing, where it is converted into grayscale to simplify the data and reduce computational complexity. After grayscale conversion, a Gaussian blur filter is applied to the image to remove noise and smoothen it. This step is important to eliminate minor variations and disturbances that may interfere with crack detection.

Following preprocessing, edge detection is performed. This stage highlights the edges and boundaries of the chain structure, making it easier to identify irregularities. Cracks on the chain surface appear as discontinuities or abnormal patterns in the detected edges.

In the crack identification stage, the system analyzes these irregular edge patterns to determine the presence of cracks. Any abnormal deviation from the normal structure of the chain is identified as a potential defect.

Finally, the output display stage presents the processed image with marked crack locations in real time. The system may also generate outputs such as processed images with crack markings, detection logs, and inspection reports, which can be used for further analysis and documentation. This step-by-step methodology ensures accurate, efficient, and continuous detection of cracks in industrial environments.

3.1 Overview of the Proposed System

The proposed system is a real-time automated crack detection solution designed for inspecting interlocked metal chains in industrial environments. It operates using a non-contact inspection mechanism, which means that the system does not physically interact with the chains during the inspection process. Instead, it relies on visual data captured through an industrial camera and processed using image processing techniques.

The system is designed to work continuously as the chains move along a conveyor belt. The camera captures high-quality images of the moving chains under controlled lighting conditions, ensuring that the surface details are clearly visible. These images are then transferred to a processing unit, where various image processing operations are performed to detect cracks.

The system integrates both hardware and software components to achieve its functionality. The hardware components include an industrial camera, camera lens, lighting system, conveyor belt, frame and mounting structure, processing unit, display unit, power supply, and connecting cables. The software component is developed using Python programming and utilizes tools such as OpenCV for image processing and analysis.

The main objective of the system is to provide accurate and real-time detection of surface cracks, improve inspection accuracy, and reduce manual effort. It is capable of identifying small and early-stage cracks that are often missed in manual inspection. The system also provides visual feedback by displaying the detected crack locations, allowing operators to quickly identify defective chains. Additionally, it supports data storage and reporting, which helps in maintaining inspection records and improving quality control processes.

3.2 Detailed Description Of The System

The proposed system for real-time automatic crack detection in interlocked metal chains is designed as an integrated combination of hardware and software components that work together to perform continuous inspection in industrial environments. The system operates using a non-contact approach, where images of the metal chains are captured and analyzed without physically interfering with the production process. This allows the system to function efficiently even when the chains are moving continuously on a conveyor belt.

The system begins with the image acquisition stage, where an industrial camera is used to capture live images of the metal chains. The camera is positioned in such a way that it focuses clearly on the chain links as they pass through the inspection area. To ensure that the captured images are of high quality, a proper lighting system is provided. This lighting setup helps in maintaining consistent illumination, which is essential for clearly highlighting the surface features of the chains. The camera, along with the lighting system, is mounted on a stable frame and mounting structure to avoid disturbances and ensure accurate image capture during operation.

Once the images are captured, they are sent to the processing unit, which acts as the central component of the system. The processing unit is responsible for executing the image processing operations required for crack detection. The first step in processing is the selection of the region of interest. In this step, only the part of the image that contains the metal chain is selected, while unnecessary background elements such as the conveyor belt and surrounding environment are excluded. This helps in focusing the analysis on the relevant area and improves the efficiency of the system.

After selecting the region of interest, the system performs preprocessing on the image. The image is first converted into grayscale, which simplifies the data by removing color information while retaining important structural details of the chain. Following this, Gaussian blur is applied to the image to remove noise and smooth the image. This step helps in reducing unwanted variations and ensures that the important features of the chain surface are clearly visible for further analysis.

The next stage in the system is edge detection, where algorithms such as the Canny edge detection method are applied. This step identifies the edges and boundaries of the chain links by highlighting the transitions in the image. These edges represent the structure of the chain, and any irregularities or disruptions in these edges may indicate the presence of cracks.

In the crack identification stage, the system analyzes the detected edges to identify abnormal patterns. Cracks appear as discontinuities or irregularities in the structure of the chain, and the system detects these deviations to determine the presence of defects. Once a crack is identified, its location is marked on the image so that it can be easily recognized.

The final stage of the system is the output display. The processed image, along with the highlighted crack locations, is displayed in real time on the display unit. This allows operators to monitor the condition of the chains continuously and take necessary actions when defects are detected. In addition to visual output, the system may also generate detection logs and inspection reports, which can be stored for future reference and analysis.

The entire system is powered by a suitable power supply and connected through cables and connectors to ensure proper communication between components. All hardware elements, including the camera, lighting system, conveyor setup, processing unit, and display unit, work together in coordination to ensure smooth and continuous operation. By integrating these components with image processing techniques, the system provides an efficient, accurate, and reliable solution for detecting surface cracks in metal chains in real time.

3.3 Block Diagram

3.3.1 Hardware Design

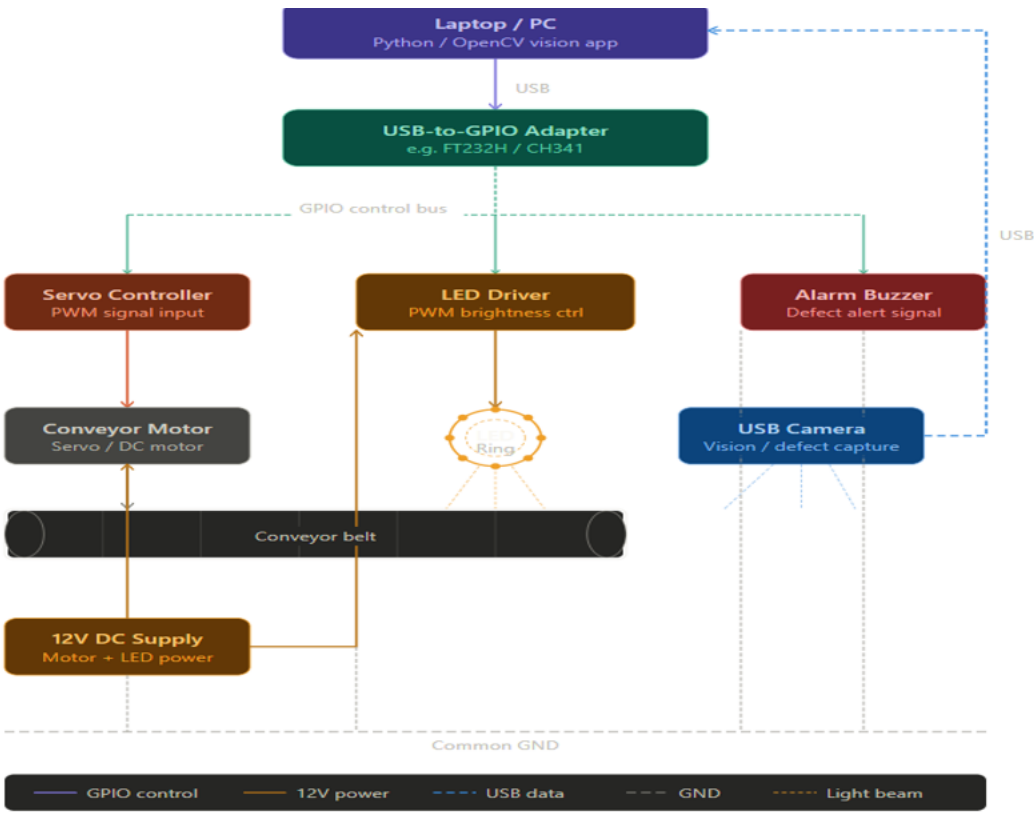

Figure 3.1: System Prototype Setup

The Figure represents an automated visual inspection system where a computer running a Python/OpenCV application controls and monitors a conveyor setup. A USB-to-GPIO adapter connects the PC to hardware components, enabling control of a conveyor motor via a servo controller, an LED lighting system for proper illumination, and an alarm buzzer for defect alerts. A USB camera captures images of objects on the moving conveyor belt, which are processed by the computer to detect defects. All components are powered by a 12V supply and share a common ground, forming an integrated system for real-time inspection and automation

Figure 3.2: Micro DC Geared Motor

The Figure shows a Micro DC Geared Motor which is a compact device that combines a high-speed brushed DC motor with a reduction gearbox to deliver high torque at lower speeds, making it suitable for applications requiring controlled motion. It operates on the principle of electromagnetism, where electric current in the armature interacts with permanent magnets to produce rotation, which is then reduced in speed and amplified in torque by the gearbox. With typical operating voltages between 3V and 12V and varying speeds depending on gear ratios, it offers features such as easy control, reversibility, and a high torque-to-size ratio. In projects, it is commonly used as an actuator for driving wheels, conveyor mechanisms, or automated systems where precise and powerful movement is needed in a limited space.

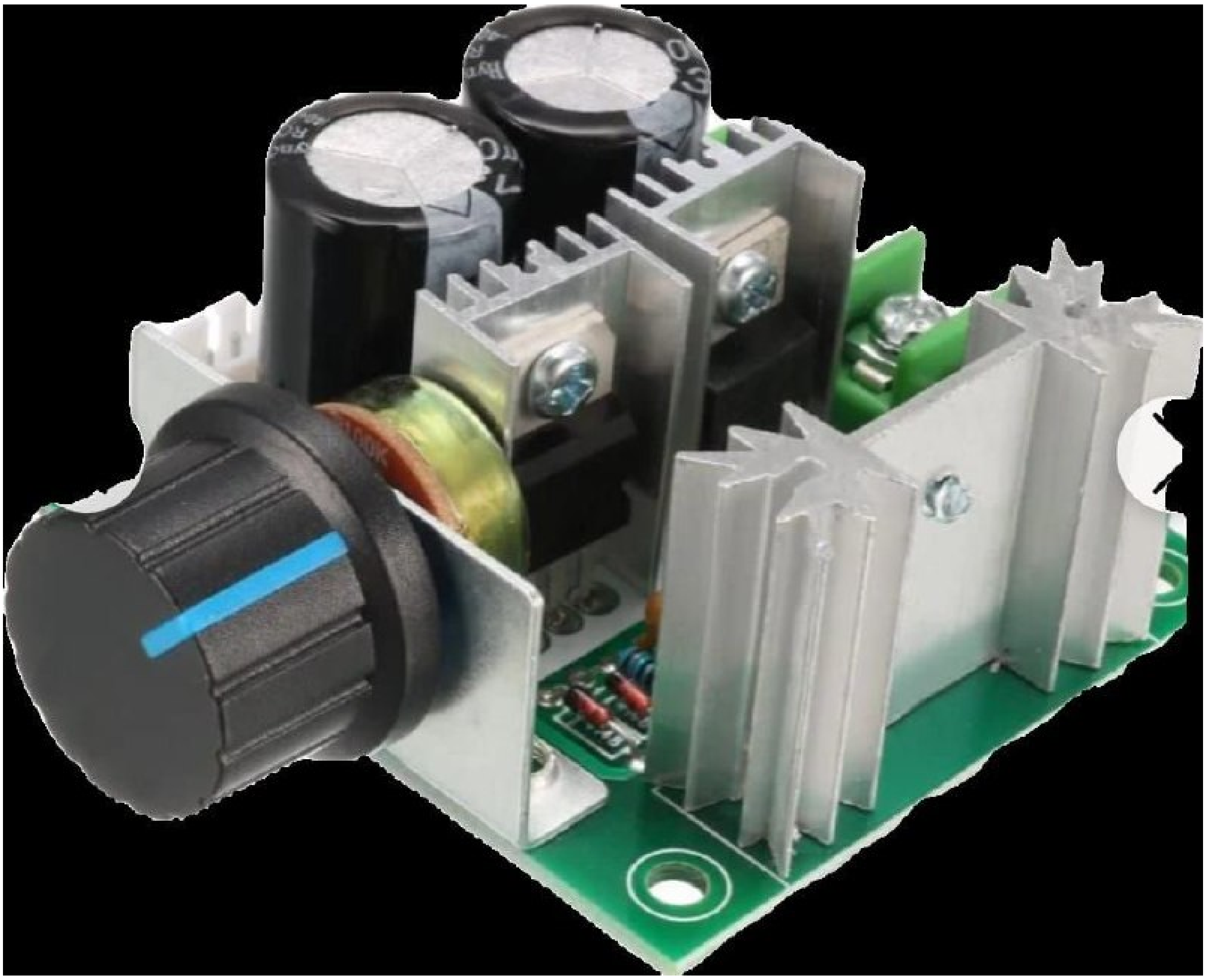

Figure 3.3: PWM DC Motor Speed Controller

The figure shows a PWM DC Motor Speed Controller which regulates motor speed efficiently by using pulse width modulation, where the power supply is rapidly switched on and off to vary the average voltage delivered to the motor without significant energy loss. Instead of reducing voltage directly, it adjusts the duty cycle of the signal, allowing the motor to maintain torque even at lower speeds. Typically supporting input voltages between 6V and 28V and currents up to several amps, it includes components such as a potentiometer for manual control, MOSFETs for switching, capacitors for smoothing, and terminals for connections. This controller is highly efficient, reduces heat loss, and simplifies implementation in projects by enabling smooth and precise speed control without requiring complex programming.

Figure 3.4: 12V AC-DC SMPS Module

The figure shows a 12V AC-DC SMPS Module which is a compact and efficient power supply unit that converts high-voltage AC mains electricity into a stable 12V DC output using high-frequency switching technology. It operates through stages including rectification, high-frequency switching, transformation via a ferrite transformer, and output regulation, resulting in high efficiency and reduced size compared to traditional power supplies. With features such as over-voltage, over-current, and short-circuit protection, it ensures safe and reliable operation. Key components include a bridge rectifier, transformer, capacitors, optocoupler, and EMI filter, all working together to provide a clean and regulated output. In projects, it serves as the main power source, supplying energy to components like motors and controllers directly from a wall outlet.

Figure 3.5: Arducam USB Camera Module

The figure shows an Arducam USB Camera Module which is a high-performance imaging device designed for embedded and industrial applications, integrating a CMOS sensor, image signal processor, and USB interface for easy connectivity. It captures images by converting light into electrical signals, processes them internally for clarity and color balance, and transmits the data via a standard USB interface using UVC protocol, making it compatible with systems like OpenCV. With features such as high frame rates, wide-angle lenses, manual focus, and compact design, it is ideal for real-time monitoring tasks. In projects, it functions as the visual input system, capturing detailed images for tasks like crack or defect detection, ensuring accurate analysis even in dynamic conditions

The system mainly consists of an industrial camera, lighting system, conveyor mechanism, processing unit, display unit, and supporting components such as power supply, cables, and mounting structures. These components are arranged in a way that enables continuous inspection of metal chains in real time.

The process begins with the motor and conveyor system, which is responsible for moving the metal chains through the inspection area. The conveyor uses a roller drive unit to ensure smooth and consistent motion of the chains. This controlled movement is important for capturing stable and clear images.

An industrial camera is positioned above or near the conveyor belt to capture live images of the chains. The camera is supported by a proper lighting system, which ensures that the chain surface is clearly visible and free from shadows or distortions. Both the camera and lighting system are mounted using a frame and mounting structure, which provides stability and maintains a fixed position during operation.

The captured images are sent to the processing unit, which performs all the image processing operations. This unit can be a computer or system capable of running the required software. The processed results are then sent to the output interface, where the final results are displayed.

The visual display unit shows the processed images with live defect overlays, allowing operators to easily identify cracks in real time. All components are powered using a suitable power supply, and proper cables and connectors are used to ensure reliable connections between devices. The hardware design ensures smooth data flow from image capture to output display.

3.3.2 Firmware Architecture

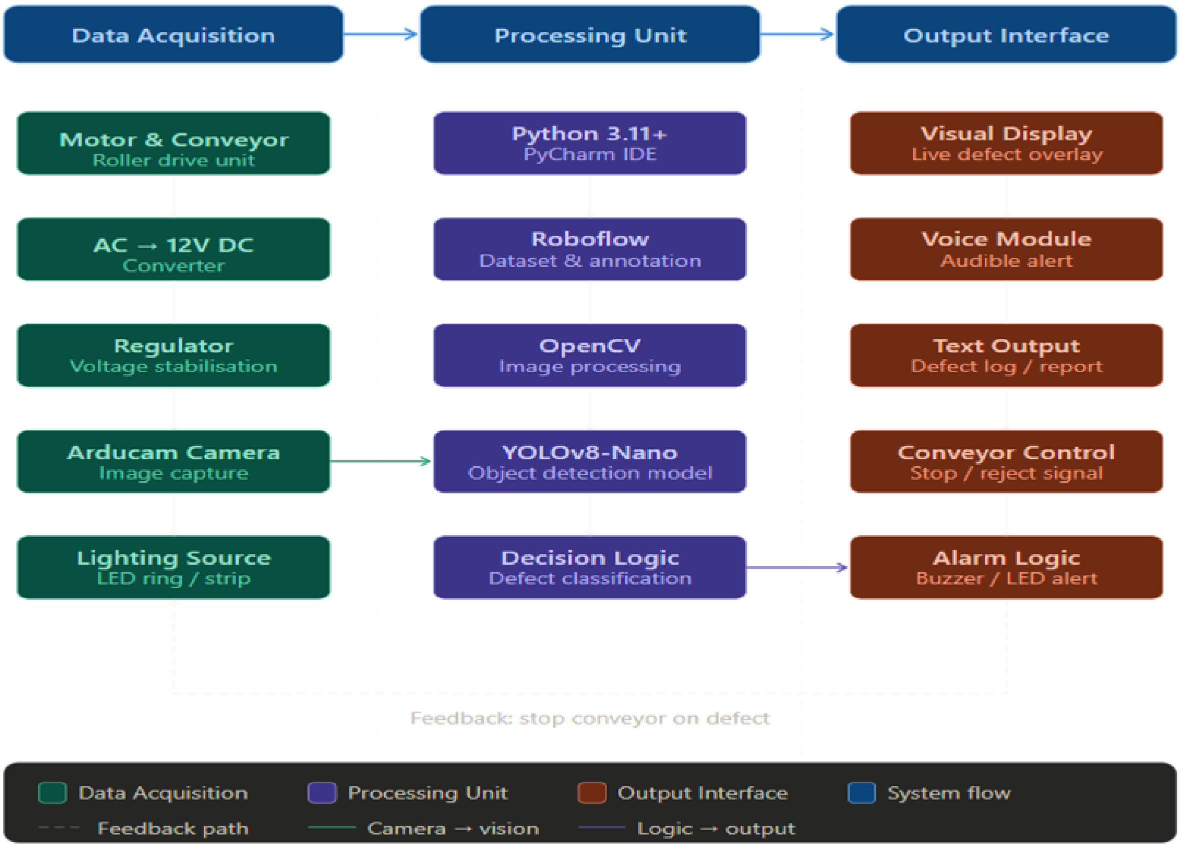

Figure 3.6: System Block Diagram

The Figure 3.6 shows the system block diagram which represents an automated defect detection system organized into three main stages: data acquisition, processing, and output interface. In the data acquisition stage, components such as the motor-driven conveyor, power supply (AC to 12V DC converter and regulator), Arducam camera, and LED lighting source work together to capture clear images of objects on the conveyor. These images are then sent to the processing unit, where tools like Python (PyCharm), Roboflow for dataset handling, OpenCV for image processing, and a YOLOv8-Nano model are used to detect and classify defects through decision logic. Finally, in the output interface stage, the system provides results through a visual display, voice alerts, and text reports, while also controlling the conveyor (stop/reject mechanism) and triggering alarms such as buzzers or LEDs when defects are identified.

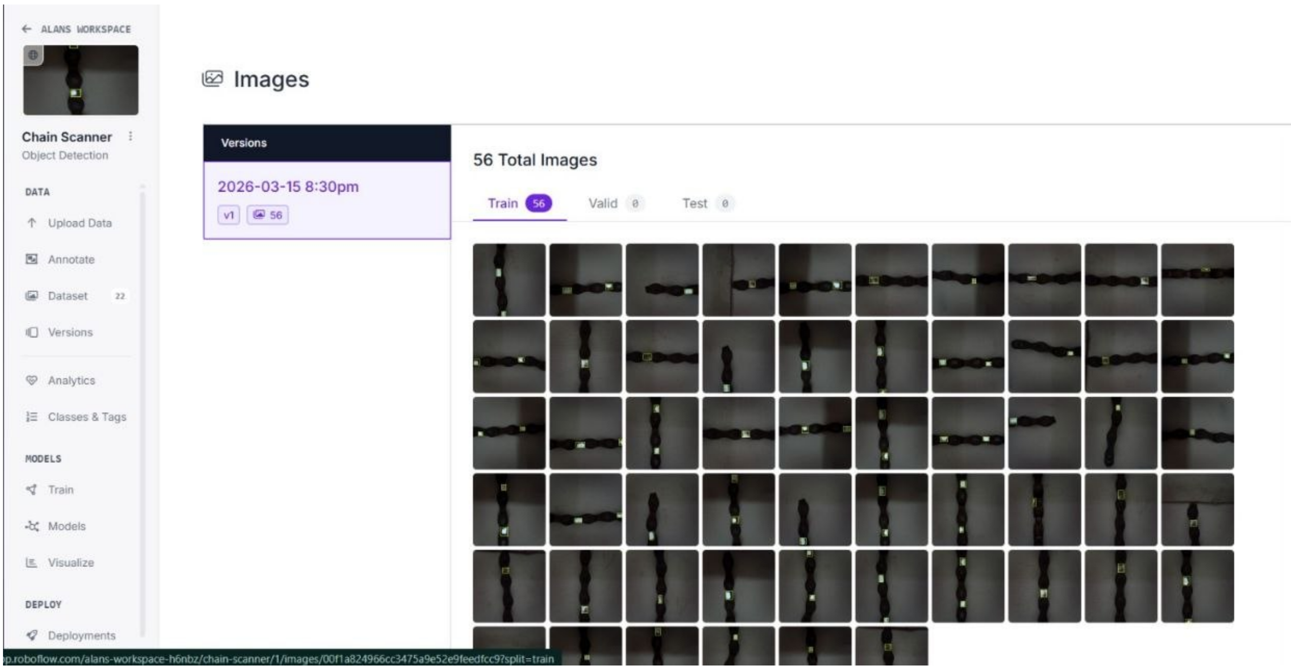

Figure 3.7: Dataset Interface

The Figure 3.7 shows a dataset interface containing multiple captured images of the chain, organized for training the crack detection model, with details such as total image count and dataset version displayed.

The system does not implement a traditional firmware architecture involving microcontrollers or embedded systems. Instead, the entire operation of the system is controlled through a software-based architecture running on a processing unit. This processing unit executes all the required functions using a high-level programming environment, specifically Python 3.8+ or Python 3.11+ with development tools such as PyCharm or Visual Studio Code, as mentioned in the software requirements.

The functional behavior of the system is defined through a sequence of software operations that handle image input, processing, and output generation. The first functional component in this architecture is the video input stage, where the software accepts a live video stream from the industrial camera. This forms the input layer of the system, and it continuously feeds image frames into the processing pipeline. The software is designed to handle real-time data input, ensuring that images are captured and processed without delay.

The next component in the architecture is the region of interest selection module. In this stage, the software allows the system to focus only on the portion of the image that contains the metal chain. By excluding unnecessary background elements such as the conveyor belt or surrounding area, the system improves processing efficiency and reduces computational load. This step is essential for ensuring that only relevant data is analyzed in subsequent stages.

Following this, the preprocessing module performs initial transformations on the input image. The software converts the captured frames into grayscale format, simplifying the image data while retaining important structural information. After grayscale conversion, Gaussian blur is applied to reduce noise and smooth the image. These preprocessing steps are implemented as part of the software logic and prepare the image for accurate analysis in the later stages.

The core of the firmware architecture lies in the crack detection module. In this stage, the software applies edge detection techniques such as the Canny edge detection algorithm to identify boundaries and edges in the image. The system then analyzes these edges to detect irregularities that may indicate the presence of cracks. This process is executed continuously for each frame received from the camera, enabling real-time detection of defects.

The final component of the architecture is the output interface module. Once cracks are identified, the software generates visual output by marking the detected crack locations on the image. These processed images are displayed on the visual display unit as live defect overlays. In addition to visual output, the system may also generate detection logs and inspection reports, which can be stored for further analysis.

Overall, the firmware architecture of the system is implemented as a sequence of interconnected software modules, each responsible for a specific function in the crack detection process. The architecture follows a structured flow from image acquisition to output generation, ensuring efficient and continuous operation. Since all processing is handled through software on the processing unit, the system does not require separate embedded firmware, and the entire functionality is achieved through programmed logic and image processing techniques.

3.3.3 Data Post-processing

The data post-processing in the system is based on the outputs generated after crack detection. Once the system processes the images and identifies cracks, it produces results in the form of processed images, detection logs, and inspection reports. These outputs serve as the basis for further analysis and record-keeping.

The processed images contain visual markings indicating the location of detected cracks. These images can be used for verification and documentation purposes. Detection logs may include information about when defects were identified, allowing tracking of inspection results over time. Inspection reports can be generated to summarize the findings, which can be useful for maintenance planning and quality control.

The system also mentions storing inspection data for analysis, which indicates that the results are not only displayed in real time but can also be saved for future reference. This helps in maintaining a record of defects and supports better decision-making in industrial operations.

The data post-processing is limited to storing, displaying, and documenting the detection results. This ensures that the system remains simple, efficient, and focused on real-time crack detection and reporting.

3.4 Flowchart

The flowchart explains the working of an automated crack detection system in a clear step-by-step manner. It begins with the start of the system, followed by the execution stage where all necessary components are initialized. In this setup phase, the trained YOLOv8 model is loaded, the camera is connected to capture live video, and the voice engine is prepared for audio output. At the same time, a variable is set to assume that the chain is safe at the beginning.

After initialization, the system starts capturing live frames from the camera continuously. Each frame acts like an image input that is sent to the AI model for processing. This continuous capturing ensures that the system monitors the chain in real time without missing any part of it. The loop keeps running as long as the system is active.

Once a frame is captured, it is passed to the AI inference stage where the YOLOv8

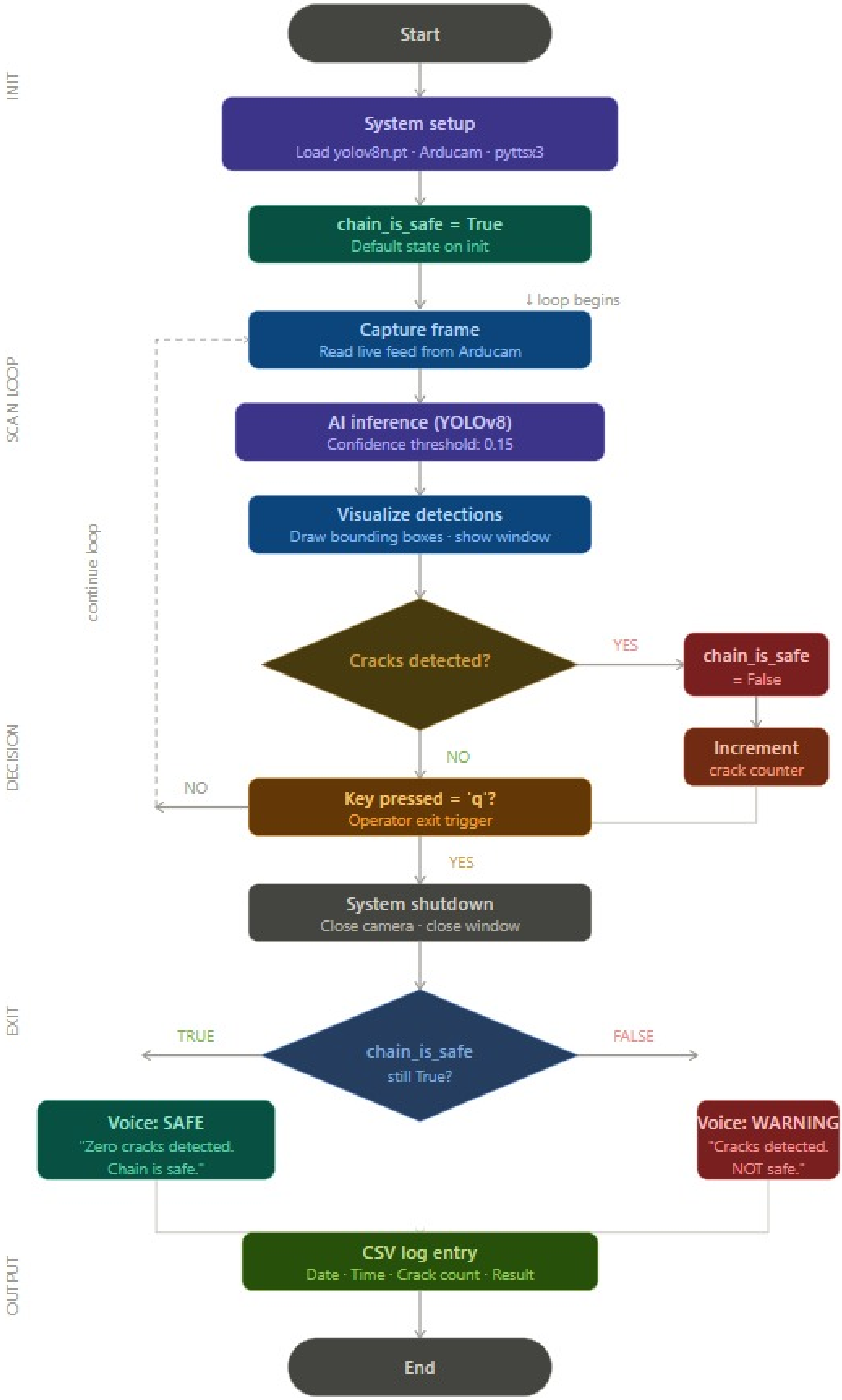

Figure 3.8: Flowchart Of the System

model analyzes the image to detect cracks. The model checks for defects based on a confidence level and identifies whether any crack is present. After detection, the system displays the result by drawing bounding boxes around the detected cracks and showing it on the screen. This helps the user visually understand where the defect is located.

The system then checks if any cracks have been detected. If no cracks are found, the system simply continues capturing and processing the next frames. However, if cracks are detected, the system changes the safety status of the chain to unsafe. This decision is important because it determines the final result of the inspection. The system continues this process until the operator decides to stop it.

When the operator stops the system, the camera is turned off and the system moves to the final stage. Here, it checks the safety status of the chain. If no cracks were detected, the system announces that the chain is safe. If cracks were detected, it gives a warning that the chain is not safe. Finally, all the details such as date, time, number of cracks, and result are stored in a file for future reference, and the process ends.

3.5 Summary

The system design is based on the integration of hardware and software components that work together to perform continuous inspection in industrial environments. The hardware design includes essential components such as the industrial camera, lighting system, conveyor mechanism, processing unit, and display unit, all arranged to ensure smooth operation and accurate image capture.

The firmware architecture of the system is implemented entirely through software, where all functionalities are executed using Python and image processing libraries. The architecture follows a structured sequence of operations, including image acquisition, region of interest selection, preprocessing, edge detection, crack identification, and output display. This ensures efficient real-time processing and accurate detection of surface cracks.

The communication within the system is achieved through direct data transfer between components, where image data flows from the camera to the processing unit and then to the display interface. The system does not use any specific communication protocol but relies on simple and effective connections for data handling. The visualization of results is carried out through a visual display unit that shows processed images with live defect overlays, enabling easy identification of cracks.

Finally, the chapter discussed data post-processing, where the system generates outputs such as processed images, detection logs, and inspection reports. These outputs can be stored and used for analysis, quality control, and maintenance purposes. Overall, the chapter highlights how the proposed system provides an efficient, reliable, and real-time solution for detecting cracks in metal chains, improving inspection accuracy and supporting industrial safety and productivity.

Chapter 4

Results and Discussions

4.1 Results of the Proposed System

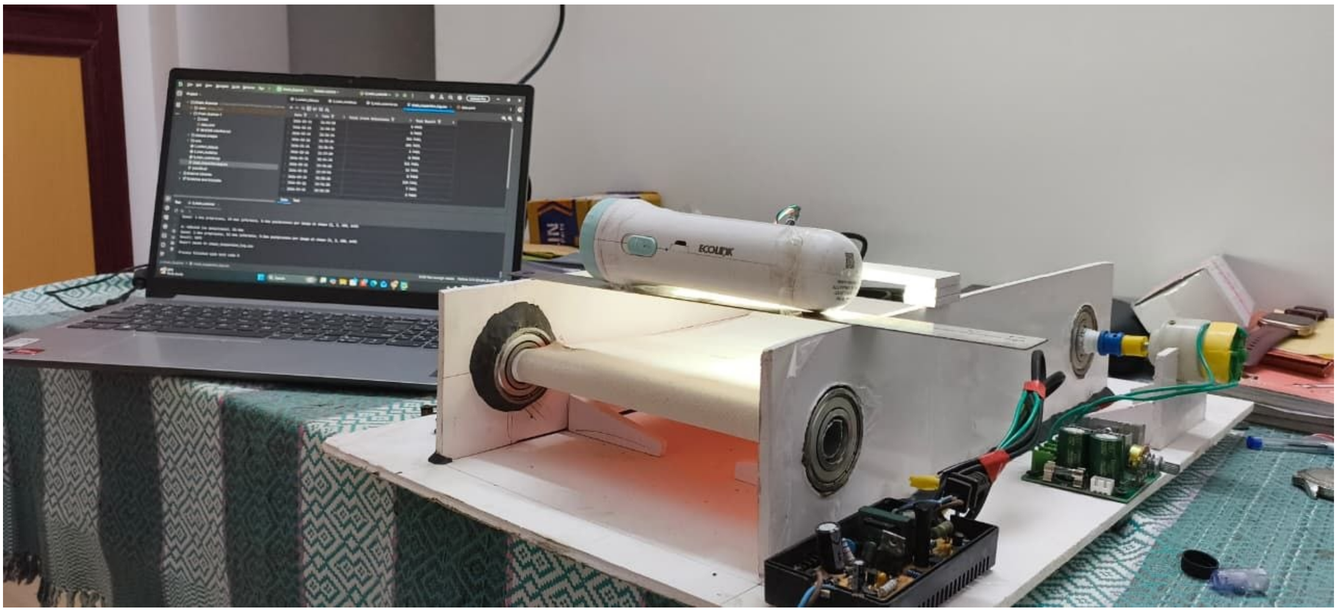

Figure 4.1: Prototype of the System

The Figure shows a prototype conveyor belt system integrated with a vision-based crack detection setup. A camera mounted above the moving belt captures images of the material surface, while a laptop connected to the system runs the detection algorithm and processes the data in real time. Electronic components and a motorized mechanism are used to drive the belt, enabling continuous inspection of objects for defects.

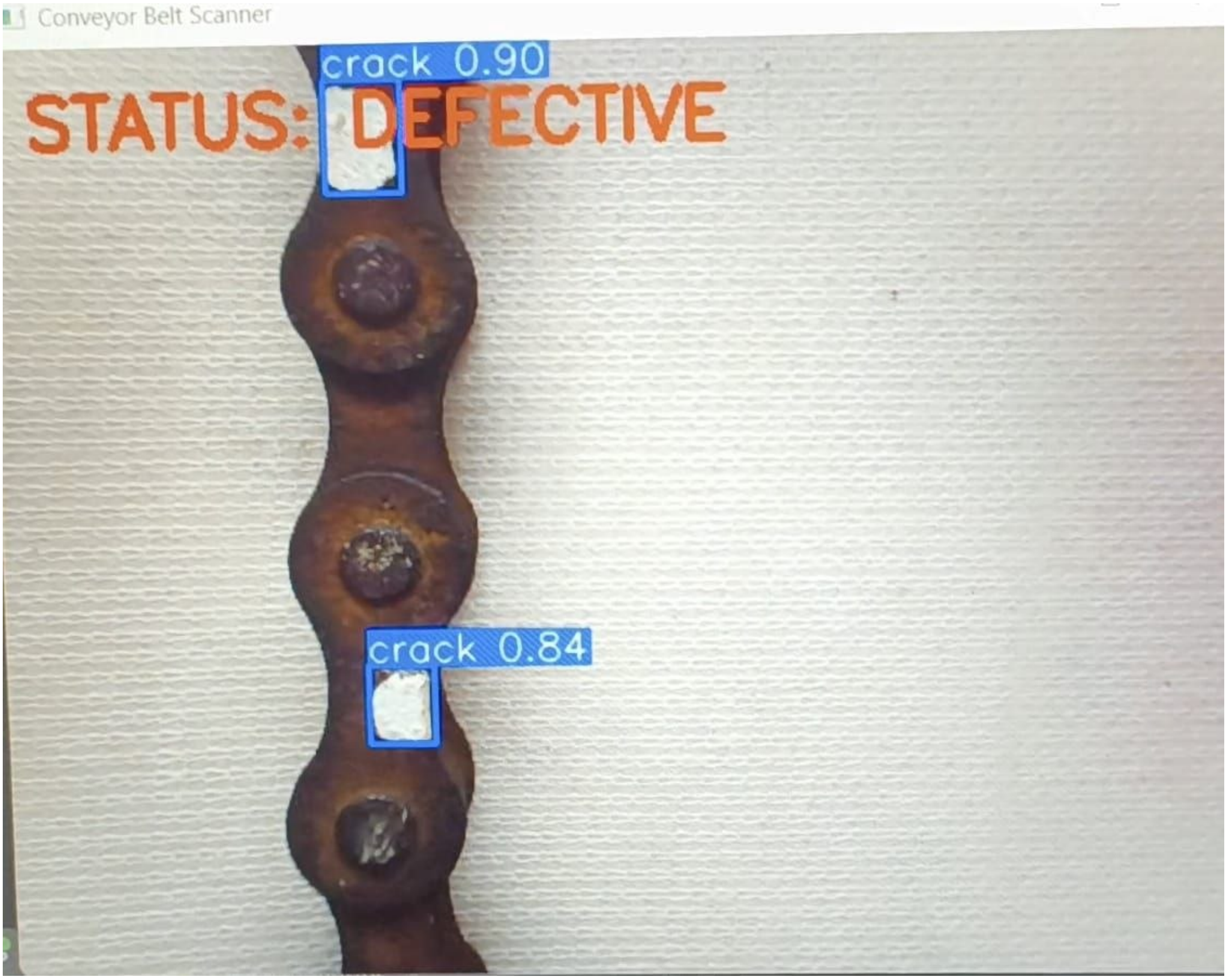

Figure 4.2: Real-Time Crack Detection Output

The Figure shows a real-time crack detection output where defects on a metal chain are identified. The system labels the inspected object as “DEFECTIVE”, indicating the presence of cracks detected during analysis.

	Date	Time	Total Crack Detections	Test Result
1	2026-03-15	21:03:50	0	PASS
2	2026-03-15	21:08:15	0	PASS
3	2026-03-15	22:31:16	351	FAIL
4	2026-03-15	22:35:24	691	FAIL
5	2026-03-15	22:36:26	1	FAIL
6	2026-03-15	22:37:39	0	PASS
7	2026-03-15	22:44:18	311	FAIL
8	2026-03-15	22:45:10	12	FAIL
9	2026-03-15	22:45:56	0	PASS
10	2026-03-15	22:50:10	329	FAIL
11	2026-03-15	22:51:28	7	FAIL
12	2026-03-15	22:52:30	0	PASS
13	2026-03-16	11:02:43	647	FAIL
14	2026-03-16	11:03:46	0	PASS
15	2026-03-16	11:30:10	0	PASS
16	2026-03-16	11:30:35	0	PASS
17	2026-03-16	11:32:37	892	FAIL
18	2026-03-16	12:03:16	961	FAIL

Figure 4.3: Screenshot of Log File

The Figure shows the screenshot of a log file generated by the crack detection system, recording the date, time, total number of cracks detected, and the corresponding test result (PASS/FAIL) for each inspection instance.

The proposed real-time automatic crack detection system successfully demonstrates the ability to detect surface cracks in interlocked metal chains during continuous operation. The system captures live images of the chains using an industrial camera and processes them in real time to identify defects. The output is displayed as processed images where crack locations are clearly marked, enabling easy identification by the operator.

The system produces multiple forms of output, including visual display with crack overlays, detection logs, and inspection reports. These outputs confirm that the system is capable of performing continuous inspection without interrupting the movement of chains on the conveyor belt. The ability to process images in real time ensures that defects are identified immediately as they appear, which is a significant improvement over traditional manual inspection methods.

The results also show that the system can detect small and early-stage cracks that are often missed during manual inspection. By focusing on the region of interest and applying image processing techniques, the system effectively highlights irregularities in the chain surface. This contributes to improved inspection accuracy and reliability.

4.2 Discussion on System Performance

The performance of the system can be evaluated based on its ability to meet the objectives defined in the project. One of the key achievements is the improvement in inspection accuracy compared to manual methods. Since the system uses automated image processing, it reduces the chances of human error and ensures consistent detection results.

Another important aspect of performance is real-time operation. The system is capable of processing images continuously as the chains move on the conveyor, which makes it suitable for industrial environments where uninterrupted operation is required. This real-time capability allows immediate detection and response to defects, enhancing overall efficiency.

The system also reduces manual effort significantly. Instead of relying on human inspection, the automated system performs the detection process and provides clear visual feedback. This not only saves time but also allows operators to focus on decision-making rather than inspection.

In terms of safety, the system contributes to preventing chain failures by detecting cracks at an early stage. This helps in reducing the risk of accidents and equipment damage. The system also supports better quality control by ensuring that defective chains are identified and handled appropriately.

4.3 Output Analysis

The output of the system is primarily visual, where processed images display crack locations with clear markings. This visual representation makes it easy for operators to understand the results without requiring technical interpretation. The system also generates detection logs, which provide a record of identified defects over time.

Inspection reports generated by the system can be used for documentation and analysis. These reports help in tracking the condition of chains and planning maintenance activities. The combination of visual output and stored data enhances the usability of the system in industrial applications.

4.4 Advantages Observed

The results of the system highlight several advantages. The system improves inspection accuracy by detecting even small cracks that are difficult to identify manually. It enables real-time monitoring, allowing continuous inspection without stopping the production process. The reduction in manual effort leads to increased efficiency and productivity. Additionally, the system enhances safety by preventing unexpected failures and supports better maintenance planning through data storage and reporting.

4.5 Performance Evaluation

4.5.1 Evaluation Criteria

The performance of the proposed system is evaluated based on criteria such as detection accuracy, real-time processing capability, reliability, and efficiency. These factors are important in determining how well the system performs in an industrial environment.

Detection accuracy refers to the system's ability to correctly identify cracks on the metal chain surface. Real-time processing capability ensures that the system can analyze images continuously without delay. Reliability indicates the consistency of the system in providing correct results, while efficiency relates to reduced manual effort and faster inspection.

4.5.2 Detection Accuracy

The system demonstrates improved detection accuracy by identifying surface cracks using image processing techniques. It is capable of detecting small and early-stage cracks that are often missed during manual inspection. By focusing on the region of interest and applying preprocessing and edge detection methods, the system enhances the visibility of defects and improves accuracy.

4.5.3 Real-Time Performance

The system demonstrates improved detection accuracy by identifying surface cracks using image processing techniques. It is capable of detecting small and early-stage cracks that are often missed during manual inspection. By focusing on the region of interest and applying preprocessing and edge detection methods, the system enhances the visibility of defects and improves accuracy.

4.5.4 Reliablity and Consistency

The automated nature of the system ensures consistent performance compared to manual inspection. Since the detection process is based on programmed logic, it produces uniform results without being affected by human factors such as fatigue or inattention. This improves the reliability of the inspection process.

4.5.5 Limitatons (Based on Scope)

The system is limited to detecting surface cracks using image processing techniques. It relies on proper lighting and image quality for accurate detection. The performance may depend on the quality of captured images and the operating conditions of the system. Advanced detection methods or additional features are not included within the current scope.

Chapter 5

Conclusion

The project titled Real-Time Automatic Crack Detection System for Interlocked Metal Chains successfully presents an efficient and reliable solution for detecting surface cracks in industrial metal chains. The system addresses the major limitations of traditional manual inspection methods, which are time-consuming, prone to human error, and often ineffective in identifying small or early-stage cracks. By introducing an automated, non-contact inspection approach, the proposed system enhances the accuracy, consistency, and speed of defect detection in industrial environments.

The system integrates both hardware and software components to perform real-time inspection of metal chains as they move on a conveyor belt. The use of an industrial camera and proper lighting ensures high-quality image acquisition, while the processing unit applies image processing techniques such as region of interest selection, preprocessing, and edge detection to identify cracks. The system is capable of providing real-time visual feedback by marking crack locations on the display, enabling operators to quickly recognize and respond to defects. Additionally, the generation of processed images, detection logs, and inspection reports supports better documentation and analysis.

The results of the system demonstrate its ability to improve inspection accuracy and detect even small cracks that are often missed during manual inspection. The real-time operation of the system allows continuous monitoring without interrupting production processes, making it highly suitable for industrial applications. By reducing manual effort and minimizing human error, the system enhances efficiency and productivity in manufacturing and maintenance operations.

Furthermore, the system contributes significantly to industrial safety by enabling early detection of defects and preventing potential chain failures. This helps in reducing the risk of accidents, equipment damage, and downtime. It also improves product quality by ensuring that defective chains are identified and removed before use. The system supports preventive maintenance practices by providing inspection data that can be used for monitoring and decision-making.

In conclusion, the proposed crack detection system provides a practical and effective solution for real-time inspection of metal chains. It improves reliability, safety, and efficiency in industrial environments while addressing the shortcomings of existing inspection methods. The project lays a strong foundation for further improvements and demonstrates the importance of automation and image processing in modern industrial applications.

5.0.1 Future Scope

The proposed real-time automatic crack detection system provides a strong foundation for further improvements and enhancements in both hardware and software aspects. Based on the developments mentioned in the project, several future advancements can be implemented to improve system performance, accuracy, and industrial applicability.

One of the major areas for future improvement is in hardware and performance upgrades. The system can be enhanced by integrating edge computing devices such as dedicated processing units, which can perform high-speed image processing directly at the inspection site. This would improve the overall speed and efficiency of the system. Additionally, the use of advanced imaging techniques such as multi-spectral imaging, including infrared or ultraviolet lighting, can help in detecting internal or micro-level cracks that are not visible through standard imaging methods. The system can also be upgraded with high-speed line scan cameras to maintain detection accuracy even at higher conveyor speeds.

Another important area of future scope is the improvement of software and intelligence. The system can be enhanced by expanding the dataset used for detection to include a wider variety of defects such as rust, deformation, and wear, in addition to cracks. This would improve the versatility of the system. Further improvements can be made by upgrading the detection algorithms to more advanced models as they become available, which can increase detection accuracy and reduce false detections. The system can also be extended to support cloud-based monitoring, allowing remote access to inspection data and enabling real-time monitoring from different locations.

The project also has scope for integration with industrial automation systems to support smart factory applications. The system can be connected to control units to automatically stop the conveyor when a critical defect is detected, ensuring immediate corrective action. It can also be extended to support predictive maintenance by analyzing stored inspection data to predict potential failures before they occur. Additionally, mechanisms can be introduced to automatically separate defective chains from the production line, improving efficiency and reducing manual intervention.

Overall, the future scope of the project lies in enhancing system capabilities, improving detection accuracy, and integrating with advanced industrial technologies. These improvements will further strengthen the system's role in ensuring safety, reliability, and efficiency in industrial operations.

Chapter 6

Experimental Results

The system was tested under a controlled setup where metal chains were allowed to move continuously on a conveyor belt while being monitored by an industrial camera. The purpose of these experiments was to evaluate the system's ability to capture images, process them in real time, and accurately detect surface cracks. The results are based on the outputs generated by the system, including processed images, visual crack markings, detection logs, and inspection reports.

6.0.1 Experimental Setup

The experimental setup consists of the hardware components described in the system design. An industrial camera is positioned above the conveyor belt to capture live images of the metal chains. A proper lighting system is used to ensure clear visibility of the chain surface and to reduce shadows or distortions in the captured images. The chains are moved using a conveyor mechanism, which ensures continuous motion during inspection. The captured images are sent to the processing unit, where the image processing operations are performed using software developed in Python with the support of OpenCV libraries. The processed results are displayed on a visual display unit in real time. The setup is designed to simulate an industrial environment where continuous inspection is required without interrupting the production process.

6.0.2 Image Acquistion Results

The image acquisition stage successfully captures clear and continuous images of the moving metal chains. The use of proper lighting ensures that the chain surface is clearly visible, which is essential for accurate detection. The camera is able to capture images without motion blur under the controlled setup, providing a stable input for further processing.

The results show that the system is capable of maintaining consistent image quality during continuous operation. This consistency is important because variations in image quality can affect the accuracy of crack detection. The captured images provide sufficient detail to identify surface irregularities on the chains.

6.0.3 Preprocessing and Edge Detection Results

During preprocessing, the captured images are converted into grayscale and subjected to noise reduction techniques. This results in smoother images with reduced unwanted variations. The preprocessing stage improves the clarity of important features while minimizing disturbances that may affect detection.

The edge detection stage highlights the boundaries and structure of the chain links. The results show that edges are clearly identified, making it easier to observe irregularities on the chain surface. Cracks appear as disruptions or discontinuities in these edges, allowing the system to distinguish them from normal chain structures.

6.0.4 Crack Detection Results

The crack detection stage demonstrates the core functionality of the system. The system is able to identify surface cracks on metal chains by analyzing irregular patterns in the processed images. The detected cracks are marked clearly on the output images, making them easy to identify.

The results indicate that the system can detect small and early-stage cracks that are not easily visible during manual inspection. This improves the overall effectiveness of the inspection process. The detection is performed in real time, ensuring that defects are identified immediately as the chains move through the inspection area.

6.0.5 Output Visualization Results

The output of the system is displayed in real time on a visual display unit. The processed images include highlighted crack regions, which provide a clear indication of defect locations. This visual representation helps operators quickly understand the condition of the chains without requiring complex analysis.

In addition to visual output, the system generates detection logs and inspection reports. These outputs provide a record of detected defects and can be used for further analysis and maintenance planning. The combination of visual and stored outputs enhances the usability of the system.

6.1 Summary of Experimental Results

The experimental results confirm that the proposed system is capable of performing real-time crack detection with improved accuracy and efficiency. The system successfully captures images, processes them, and identifies defects while providing clear visual output. The results validate the effectiveness of the proposed approach in overcoming the limitations of traditional inspection methods and demonstrate its suitability for industrial use.

References

[1] Z. Cheng, F. Yang, S. Deng, F. Chen, and J. Huo, "An efficient real-time metal crack detection model based on RT-DETR," in Proc. 9th Asia-Pacific Conf. Intell. Robot Syst. (ACIRS), IEEE, Jul. 2024, pp. 220–224

[2] J. Liao, L. Yang, H. Tan, and F. Zhou, "A stable lightweight model for metal crack detection based on YOLOv5," in Proc. 7th Int. Conf. Image, Vis. Comput. (ICIVC), IEEE, Jul. 2022, pp. 123–128.

[3] Q. Wang, X. Xu, and Y. Zhao, "Defect detection algorithm for chain saw chain production line based on deep learning," in Proc. IEEE 2nd Int. Conf. Data Sci. Comput. Appl. (ICDSCA), IEEE, Oct. 2022, pp. 1401–1404

[4] K. Anitha, "Next-gen quality control using data-driven approaches and machine learning in metal production," in Proc. Int. Conf. Data Sci., Agents Artif. Intell. (IC DSAAI), IEEE, Mar. 2025, pp. 1–5.

[5] M. A. Abou-Khousa, "Detection of surface cracks in metals using microwave and millimeter-wave nondestructive testing techniques—A review," IEEE Trans. Instrum. Meas., vol. 72, pp. 1–18, Jan. 2023.

Appendix

Figure 6.1: QR Code for Project Source Code and Repository